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Do Double-Hybrid Exchange–Correlation Functionals Provide Accurate Chemical Shifts? A Benchmark Assessment for Proton NMR

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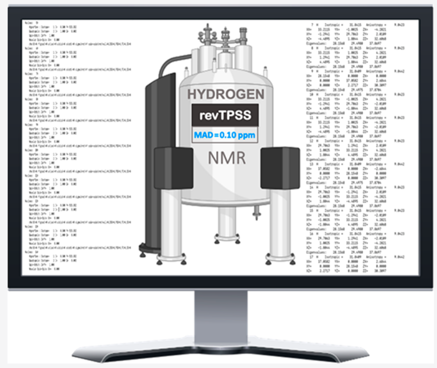
ABSTRACT: A benchmark density functional theory (DFT) study of ^{1}H NMR chemical shifts for data sets comprising 200 chemical shifts, including complex natural products, has been carried out to assess the performance of DFT methods. Two new benchmark data sets, NMRH33 and NMRH148, have been established. The meta-GGA revTPSS performs remarkably well against the NMRH33 benchmark set (mean absolute deviation (MAD), 0.10 ppm; maximum deviation (max), 0.26 ppm) with the smallest MAD of all evaluated functionals. The best-performing double-hybrid density functional (DHDF), revDSD-BLYP (MAD, 0.16 ppm; max, 0.35 ppm), performs similarly to hybrid-GGA methods (e.g., mPW1PW91/6-311G(d) (MAD, 0.15 ppm; max, 0.36 ppm)), but at a considerably higher computational cost. The results indicate that currently available double-hybrid DFT methods offer no benefit over GGA (including hybrid and meta) functionals in the calculation of ^{1}H NMR chemical shifts.



1. INTRODUCTION

The ^{1}H nuclide, with a magnetically active spin of 1/2 and natural abundance over 99.98%, had a central role in the pioneering work of Bloch's and Purcell's groups leading to the discovery of NMR spectroscopy in 1945.[1,2] The technique has become a fundamental characterization tool across the molecular sciences, whereby it is now routine to acquire ^{1}H NMR data of up to 1000 samples daily. The bottleneck remains the interpretation of spectra for complex molecules. Unsurprisingly, structural misassignments of natural products[3−11] still occur with potentially disastrous consequences for drug discovery projects.[12] In this context, quantum calculations of NMR parameters (e.g., chemical shifts and scalar coupling constants) facilitated by affordable DFT methods have developed into a complementary tool in the laborious task of structure (re)assignment.[3−11,13−17] Furthermore, ^{1}H NMR provides critical information for stereochemical assignment, which can be more discriminating than ^{13}C NMR spectroscopy.[18,19] Nonetheless, proton chemical shifts have proved particularly difficult to predict due to the location of H atoms at the periphery of molecules, where they experience the influence of other nuclei through inter- or intramolecular interactions.

For the past 20 years, the application of popular density functionals (DFs) in the hybrid-GGA class, such as B3LYP[20−22] or mPW1PW91,[23] typically in combination with a Pople basis set of double- or triple-ζ quality, has gained widespread use as a staple strategy for computing chemical shifts of organic molecules[24−26]—an approach, however, which has come under criticism for the occurrence of error compensation.[27,28] Various practical statistical,[29−33] AI-based,[34−36] and automated[37] protocols for structural assignment rely on the quality of results from popular DFT methods. Benchmarking studies of chemical shift calculations offer invaluable data to method developers and users, including the predictive capacity of DFs.[38−42] In this context, a comprehensive account by Iron[38] stands out for assessing multiple parameters including dozens of DFs and basis sets for ^{1}H and ^{13}C NMR chemical shift predictions. Their study revealed that the best results for ^{13}C NMR were obtained with long-range corrected (LC−)[43] DFs.

Double-hybrid DFs (DHDFs) are represented in the highest rank or rung 5 of Perdew's "Jacob's ladder"[44] classification of DFs and understandably have attracted substantial attention. B2PLYP,[45] the first DF developed in the class and an archetypal DHDF, combines both DF and wave function theories by substituting part of the usual DFT correlation term with an MP2-like nonlocal correlation term. DHDFs have been recommended for their superior accuracy over DFs on lower rungs of Jacob's ladder from a number of benchmark studies evaluating thermochemistry, kinetics, and noncovalent interactions.[27,46−49] Similar surveys of NMR parameters using

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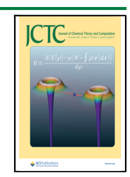

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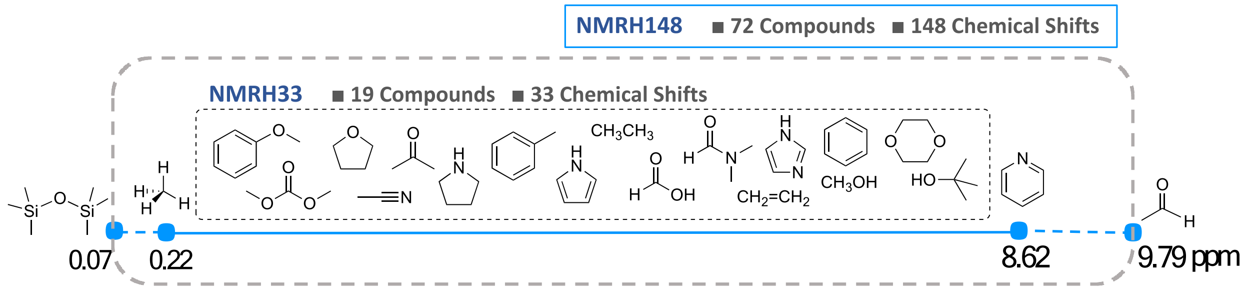

Figure 1. NMRH33 and NMRH148 data sets with ^{1}H chemical shifts ranging from 0.22 to 8.62 ppm (NMRH33) and from 0.07 to 9.79 ppm (NMRH148).

DHDFs remain scarce in the literature. In 2018, Neese et al. reported the implementation of NMR shielding tensor calculations for DHDFs in a development version of the Orca code.[50] Their study evaluated the performance of a few DHDFs (B2PLYP, B2GP-LYP, and DSD-PBEP86) among DFs from other classes for several nuclei including ^{1}H using a small data set (n = 4). Kozuch and Martin's DSD-PBEPB86[51,52] showed the best results in comparison to accurate coupled cluster and reference data. More recently, Grimme et al.[53] considered the performance of DHDFs (revDSD-PBEP86, revDSD-BLYP, B2GP-PLYP, and mPW2PLYP) for ^{29}Si shift calculations. The DHDFs tested showed "no to small improvement" when compared to other DFs in the study. To date, there has been no comprehensive investigation of the performance of DHDFs in the calculation of ^{1}H NMR chemical shifts.

In the work presented here, we have investigated a series of DFT methods across various rungs of Jacob's ladder to address the central question of *whether current DHDFs perform better than lower-rung DFT methods in predicting ^{1}H NMR chemical shifts*. Since DHDFs have a higher computational cost compared to GGA-based DFs, it is of significant interest to investigate whether the increased computational effort provides greater accuracy. ^{1}H NMR provides an ideal target property for the current study, since it is the most common NMR spectroscopy and ^{1}H is an especially relevant nuclide in structure elucidation.

For a given problem or property, the selection of a DFT method from the overwhelming number of available DFs remains largely influenced by popularity, particularly for non-experts.[27,54] The present study is focused on the central question rather than an exhaustive investigation of DFs and all possible factors that influence the calculation of chemical shifts; however, results from this work do enable recommendations of DFs for use in ^{1}H NMR calculations, which will also guide future developments for DHDFs.

2. COMPUTATIONAL METHODS

All chemical shift calculations were carried out at gas-phase M06-2X/6-311+G(2d,p) optimized geometries (without any symmetry constraints), except as otherwise indicated. Stringent criteria were imposed with "very tight" geometry convergence and "superfine" (175,974) DFT integration grid in Gaussian 09 (Rev. D.01).[55] Harmonic vibrational frequencies were calculated analytically to confirm that all optimized geometries are minima. The reference compounds have been selected to minimize effects from conformational flexibility. For the NMRH33 and NMRH148 data sets (see next section), verification that molecules are rigid or represented by only a single major conformation was carried out from a conformational search using the OPLS3[56] force field available in Macromodel.[57] For the natural products, a conformational search was carried out using Macromodel with the OPLS3 force field over an extended energy window (15 kcal/mol). Geometries of representative conformers were optimized at the M06-2X/6-311+G(2d,p) level of theory. For natural products where multiple low-energy conformations were identified, a Boltzmann distribution was determined using gas-phase ωB97M-V[27,58]/def2-TZVPPD[59,60] energies calculated at the M06-2X/6-311+G(2d,p) optimized geometries.

Chemical shifts were determined from chemical shielding constants relative to TMS calculated at the same level of theory. The gauge-independent atomic orbital (GIAO)[61] method was used throughout to compute isotropic magnetic shielding constants. Proton chemical shifts, $\delta(^{1}\mathrm{H})$, were calculated from

$$\delta(^{1}\mathrm{H}) = \sigma(^{1}\mathrm{H}_{\mathrm{reference}}) - \sigma(^{1}\mathrm{H}) + \delta(^{1}\mathrm{H}_{\mathrm{reference}})$$

where $\sigma(^{1}\mathrm{H}_{\mathrm{reference}})$ is the ^{1}H NMR shielding constant for the reference compound in $CDCl_3$, $\sigma(^{1}\mathrm{H})$ is the proton shielding constant for a given molecule of interest in the same solvent, and $\delta(^{1}\mathrm{H}_{\mathrm{reference}})$ is the chemical shift for the reference compound in $CDCl_3$. The chemical shift for the standard reference compound TMS (δ_{TMS}) in ^{1}H NMR is zero. To compare calculated chemical shifts with the experimental reference data, equivalent nuclei are averaged (e.g., shifts of three protons in a CH_3 group are averaged).

Most NMR shielding calculations were performed using Gaussian 09 (Revision D.01);[55] calculations involving MP2,[62,63] SCS-MP2,[64] double-hybrid density functionals (DHDFs), TPSSh,[65,66] and ωB97M-V[58] were performed with ORCA 4.2.1,[67] whereby the resolution of identity scheme RIJK was applied.[68−71] Dunning's cc-pVTZ basis set[72,73] was used with matching cc-pVTZ/JK and cc-pwCVQZ/C auxiliary basis sets (with cc-pVQZ/JK and cc-pwCVQZ/C auxiliary basis sets for cc-pVQZ orbital basis sets).[74−76] Jensen's pcSseg-n basis set[77] was used in conjugation with the "universal" decontracted def2-JK[74] and cc-pwCVQZ/C auxiliary basis sets.[74,75] For the evaluation of basis set convergence with the pcSseg-n series, auxiliary basis sets cc-pwCVXZ/C with one higher cardinal number were used; e.g., pcSseg-1 was paired with cc-pwCVTZ/C, until pcSseg-3 and pcSseg-4, where the cc-pwCV5Z/C auxiliary basis was employed. For MP2, SCS-MP2, and DHDFs NMR computations, core electron contribution to the correlation energy was included (all electrons are correlated).

To minimize DFT integration grid effects, a superfine (175,974) grid was employed in Gaussian 09 while in Orca the

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large "Grid7 NoFinalGrid" was selected. Implicit solvent effects were computed within the conductor-like polarizable continuum model (CPCM),[78] except as otherwise indicated.

3. DATA SETS

Two ^{1}H NMR data sets were constructed to benchmark the performance of DFs. To minimize effects from conformational flexibility, rigid molecules or those represented by a single major conformation were favored. All reference data are taken from experimental chemical shifts, with all results limited to a single solvent ($CDCl_3$) to minimize the effect of various solvents in the reference ^{1}H NMR data. The first set, labeled NMRH33, contains 33 individual proton shifts from 19 common organic compounds (Figure 1). The NMRH33 set spans over 8.42 ppm (0.22−8.62 ppm), which covers a considerable range of relevant proton shifts in organic chemistry, typically 10 ppm. An extended data set (NMRH148) with 72 compounds and 148 chemical shifts (range of 9.72 ppm) was also developed, which includes all compounds in the NMRH33 subset (for a list of additional compounds in NMRH148, see Supporting Information Figure S2).

All methods were benchmarked against the NMHR33 data set, with a select series of methods also benchmarked against the NMRH148 data set to assess the validity of the trends and conclusions that arise from the smaller data set. Geometries of all reference compounds are available in the Supporting Information. There are many variables that may affect NMR predictions, such as geometry, solvation, rotational−vibrational effects, and relativistic effects. A thorough evaluation of all variables is beyond the scope of the current work that is focused on the performance of DHDFs; however, we note that many of these effects, such as vibrational averaging, are expected to be small for ^{1}H chemical shifts[38] and are not expected to influence the conclusions from the present study.

4. RESULTS AND DISCUSSION

For an initial evaluation of the performance of DHDFs, we selected a small number of DFT methods representative of (i) common functionals used in applications and (ii) recommendations from recent benchmarking studies. The representative sample of methods (M1−M4) was assessed against the NMRH33 set of ^{1}H NMR shifts. M1 and M2 represent popular and affordable hybrid-GGA methods that are commonly used to correlate structure and spectra, including complex natural products.[31] The M3 and M4 approaches include a range-separated hybrid-GGA functional and DHDF, respectively, which have been identified from a systematic evaluation of multiple parameters (e.g., basis sets, solvation effects, and NMR calculation methods). A description of the selected methods (M1−M4) follows.

- Method 1 (M1): GIAO−CPCM/mPW1PW91/6-311G(d). Described by Goodman et al.[79] as the best method for the practical DP4 statistical analysis used in stereochemistry assignment, M1 represents a highly popular approach to calculate chemical shifts. The PCM solvation method[80] used in the original work was replaced by CPCM,[78] which shows better performance for larger molecules while offering improved results for solute−solvent interactions.[81]
- Method 2 (M2): GIAO−CPCM/ωB97X-D/6-311G(2d,2p). Hehre et al.[82] applied this approach to ^{13}C NMR shifts using a large set of natural products, while here we extend the application to ^{1}H chemical shifts. The range-separated hybrid ωB97X-D[83] has previously been used to compute ^{1}H chemical shifts.[32] The CPCM solvation model[78] was employed.
- Method 3 (M3): CSGT-COSMO/LC-TPSS/cc-pVTZ. Recommended by Iron[38] after an extensive multiparameter analysis focused on ^{13}C NMR shifts. Notably, the continuous set of gauge transformations (CSGTs)[84] and the conductor-like screening model (COSMO)[85] solvation are applied. ^{1}H NMR chemical shifts were also evaluated in the study.
- Method 4 (M4): GIAO−CPCM/DSD-PBEP86/pcSseg-2. Neese and co-workers[50] report the first evaluation of the performance of DHDFs (B2GP-PLYP and DSD-PBEP86) for a number of nuclei (^{1}H, ^{13}C, ^{15}N, ^{17}O, ^{19}F, and ^{31}P). CPCM solvation[78] was included, in addition to gas-phase calculations for comparison with the original work.

Results for the NMRH33 test set are presented in Table 1. Interestingly, M1 and M2 perform better than both M3 and the DHDF represented by M4 based on mean absolute deviation (MAD), maximum deviation (max), standard deviation (SD), and percentage mean absolute relative error (%MARE). The popular mPW1PW91/6-311G(d) method (M1: MAD, 0.15; max, 0.36 ppm) performs better than the other DFT methods considered in this preliminary evaluation. It is important to note that our approach to directly compare the calculated chemical shift to experiment differs from the approach taken by Iron,[38] who applied a linear regression to a set of calibration molecules and, therefore, produced scaled results that will vary from the present results.

Table 1. Mean Absolute Deviation (MAD), Maximum Deviation (max), and Standard Deviation (SD) in Units of ppm, and Percentage Mean Absolute Relative Error (% MARE), for Selected DFT Methods (M1−M4, See Text) for Predicting Proton NMR Chemical Shifts Using the NMRH33 Data Set

method	M1	M2	M3	M4
MAD	0.15	0.21	0.24	0.24
max	0.36	0.49	0.58	0.59
SD	0.11	0.17	0.17	0.21
%MARE	3.8	4.2	5.9	5.0

The initial results indicated that the DSD-PBEP86 (M4) double-hybrid method performed no better than other hybrid-GGA functionals (M1−M3), which prompted us to consider evaluating the performance of a greater range of DFs, while having all other parameters constant (i.e., basis sets, solvation, and NMR calculation method).

A series of DFs were subsequently evaluated against NMRH33 with an emphasis on DHDFs (B2PLYP,[45] B2GP-PLYP,[86] DSD-BLYP,[87] DSD-PBEP86,[51,52] mPW2PLYP,[88] revDOD-PBEP86,[89] revDSD-BLYP,[89] and revDSD-PBEP86[89]) and members of the TPSS family (LC-TPSS,[43,66] revTPSS,[66,90] TPSS,[66] and TPSSh[65,66]) that have been identified to perform relatively well and have been frequently evaluated in recent studies.[38,50,53] The hybrid-GGA B3LYP[20−22] and mPW1PW91[23] (rung 4) functionals were also included for comparison. All electrons were correlated in all DHDF NMR calculations. Test calculations with a frozen

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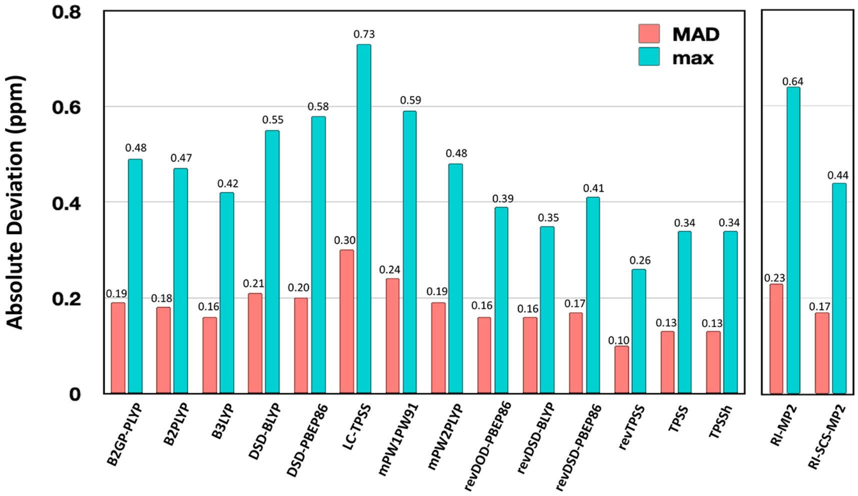

Figure 2. Results for NMRH33 data set: mean absolute deviation (MAD) and maximum deviation (max) in ppm for all tested density functionals with the cc-pVTZ basis sets. RI-MP2 and RI-SCS-MP2 methods utilized the cc-pVQZ basis set.

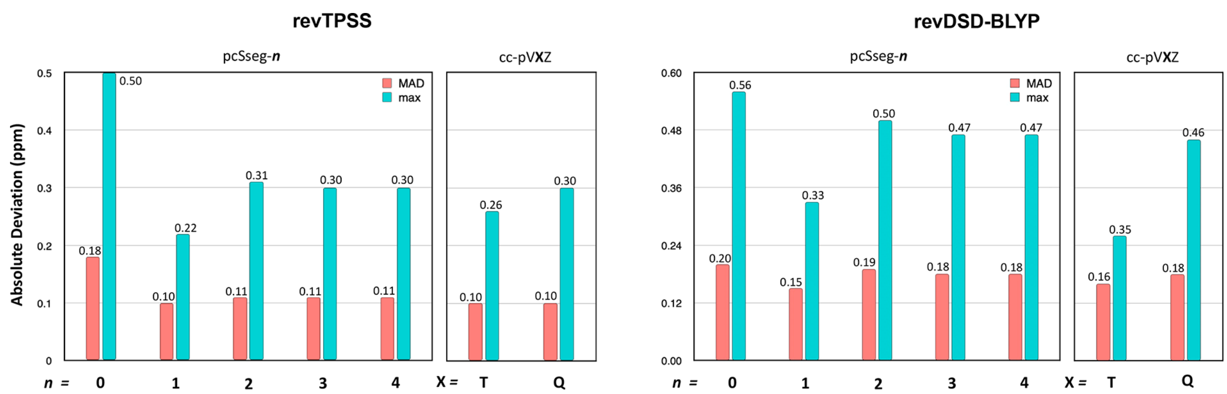

Figure 3. Mean absolute deviation (MAD) and maximum deviation (max) in ppm for revTPSS and revDSD-BLYP density functionals with pcSseg-*n* and cc-pVXZ basis sets.

core approach yielded only small numerical differences (see Supporting Information).

Two triple-ζ basis sets were considered for this evaluation: Jensen's segmented contracted pcSseg-2 basis set optimized for nuclear magnetic shielding[77] and Dunning's correlation consistent cc-pVTZ basis set.[72] Iron[38] recommended the use of Dunning's basis sets, while Jensen's sets have been reported to provide reliable results with doubled-hybrid density functionals.[50,53] We were particularly interested in the performance of the Dunning basis sets as this has not been investigated with DHDFs for NMR calculations.

Statistical results with the cc-pVTZ basis set are illustrated in Figure 2. Individual results are collated in the Supporting Information (Table S2). In general, the cc-pVTZ basis set yielded smaller MADs than the pcSseg-2 basis set. The popular hybrid-GGA B3LYP functional (MAD, 0.16 ppm; max, 0.42 ppm) performed reasonably well, with results comparable to the best-performing DHDF (revDSD-BLYP: MAD, 0.16 ppm; max, 0.35 ppm). The latter yields a smaller maximum deviation, but at a substantially higher computational cost. The hybrid-GGA mPW1PW91 with pcSeg-2 did not perform as well (MAD, 0.24 ppm; max, 0.59 ppm), which may be compared to results with mPW1PW91/6-311G(d) (MAD, 0.15 ppm; max, 0.36 ppm) that points toward error cancelation between the DFT method and the basis set.[49]

In general, the DHDFs performed similarly with the pcSseg-2 and cc-pVTZ basis sets, with the cc-pVTZ basis sets yielding slightly smaller MADs, maximum deviation, %MARE, and standard deviation. In order to investigate the importance of the MP2 contribution in the DHDF we calculated chemical shifts with second-order Møller−Plesset perturbation theory (MP2)[62,63] and the spin-component-scaled MP2 (SCS-MP2)[64] methods employing the resolution of identity (RI) approximation. In both cases, the cc-pVQZ basis set was used to provide results closer to the basis set limit (for details, see Table S4). RI-MP2/cc-pVQZ performs similarly to mPW1PW91/cc-pVTZ (MAD, 0.23 ppm; max, 0.64), with a MAD that is larger than any of the DHDFs considered in this study. RI-SCS-MP2/cc-pVQZ (MAD, 0.17 ppm; max, 0.44) performs significantly better than RI-MP2, but only slightly better than the best-performing DHDFs, revDOD-PBEP86 and revDSD-BLYP. The results suggest that inclusion of MP2 correlation by itself is insufficient since MP2 systematically underestimates correlation energy, while accounting for higher-order correlation with the empirical SCS-MP2 approach reduces the deviation from experimental values.

An interesting outcome is that the DHDFs only performed at a level equivalent to the hybrid-GGA B3LYP functional (rung 4), both in terms of MAD and maximum deviation, despite the increased computational cost of double hybrids. Importantly, the TPSS family exhibited some of the best

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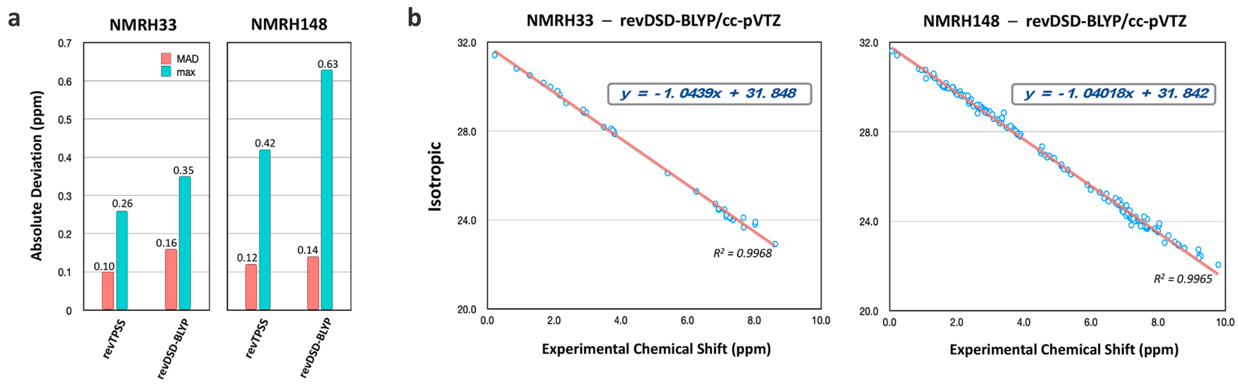

Figure 4. Evaluation of the data sets. (a) Mean absolute deviation (MAD) and maximum deviation (max) in ppm for revTPSS and revDSD-BLYP using NMRH33 and NMRH148 data sets; (b) isotropic (revDSD-BLYP/cc-pVTZ) versus experimental chemical shift plots for both data sets.

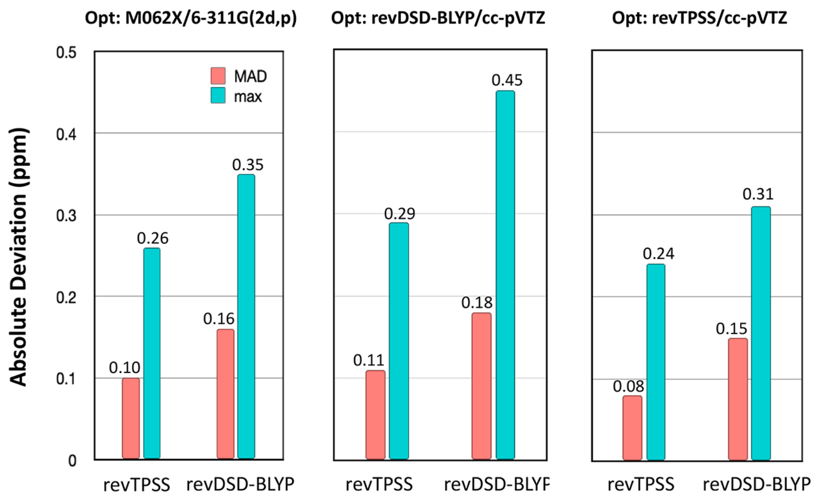

Figure 5. Mean absolute deviation (MAD) and maximum deviation (max) in ppm for revTPSS and revDSD-BLYP/cc-pVTZ using geometries optimized (Opt) at different levels of theory.

overall performance with the exception of LC-TPSS (MAD, 0.30 ppm; max, 0.73 ppm).[38] The excellent performance of the meta-GGA revTPSS (rung 3) functional (MAD, 0.10 ppm; max, 0.26) is a notable outcome, which has been overlooked in previous studies.

Basis set convergence was investigated with the revTPSS and revDSD-BLYP methods. We considered the entire family of pcSseg-n (n = 0−4) basis sets,[77] which extends from single-ζ to 5Z (Figure 3, see Supporting Information Table S6 for individual results). There is no significant improvement in statistics beyond triple-ζ (pcSseg-2) with both revTPSS and revDSD-BLYP. Results with pcSseg-1 (double-ζ) yielded the smallest MAD and maximum deviations, which occur due to well-known error cancellation effects as indicated by results with larger basis sets.[27,28] We also considered Dunning's cc-pVXZ (X = T, Q) basis sets. Agreement with experiment was better with the TZ basis set, which again may be ascribed to error cancellation effects. Results with both Dunning and Jensen basis sets indicate that the TZ basis sets yield chemical shifts that are reasonably close to basis set convergence in terms of MAD (Figure 3). Calculations with QZ basis sets did not result in any major improvement, despite the significant increase in computational cost.

Interestingly, revTPSS/cc-pVTZ (MAD, 0.10 ppm; max, 0.26 ppm) was ranked the overall best method, outperforming results with all pcSseg-n basis sets—a result in agreement with the comprehensive NMR benchmark study of Iron that did not include DHDFs.[38] Likewise, revDSD-BLYP/cc-pVTZ was ranked the best DHDF (MAD, 0.16 ppm; max, 0.35 ppm).

Calculations applying the 4.5-fold larger NMRH148 set provide conclusions that are consistent with the NMRH33 data set, which serves to validate our "express" approach with the smaller NMRH33 data set. The best DHDF (revDSD-BLYP: MAD, 0.16) and meta-GGA DF (revTPSS: MAD, 0.12) perform similarly well for both data sets (Figure 4a). While the MADs remain similar between the two benchmark sets, the maximum deviations (max) consistently increase for NMRH148 with the inclusion of more complex structures, a greater range of heteroatoms, and more difficult cases. The difference between the data sets is readily visualized by comparing plots of isotropic values against experimental chemical shifts. The NMRH148 set not only provides an extended range of 0.07−9.79 ppm as compared to NMRH33, but also includes more data points with intermediate chemical shifts arising from greater chemical diversity. Nonetheless, the gradients in both graphs are very similar (Figure 4b).

Calculated chemical shifts rely on the calculated nuclear magnetic shielding for both the sample and reference molecules, for which we used TMS as the reference for ^{1}H chemical shifts. To assess whether results were dependent on the calculated shielding of TMS, we generated a set of results using methane as the ^{1}H reference using the NMRH148 data

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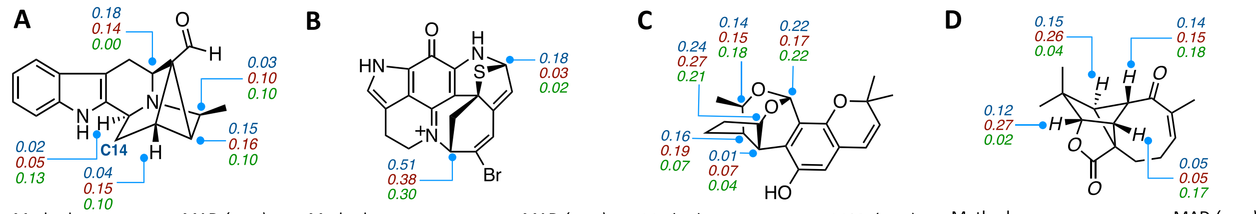
TOTAL (*n=52*)	
B3LYP or mPW1PW91 + Pople TZ	0.16 (*0.70*)
revDSD-BLYP/cc-pVTZ	0.18 (*0.68*)
revTPSS/cc-pVTZ	0.11 (*0.46*)



A

Method	MAD (*max*)
B3LYP/6-311G(d,p)	0.13 (*0.30*)
revDSD-BLYP/cc-pVTZ	0.19 (*0.58*)
revTPSS/cc-pVTZ	0.10 (*0.17*)

B

Method	MAD (*max*)
mPW1PW91/6-311+G(d,p)*	0.16 (*0.51*)
revDSD-BLYP/cc-pVTZ	0.14 (*0.38*)
revTPSS/cc-pVTZ	0.13 (*0.30*)

*corrected

C

Method	MAD (*max*)
mPW1PW91/6-311G(d)	0.22 (*0.70*)
revDSD-BLYP/cc-pVTZ	0.22 (*0.68*)
revTPSS/cc-pVTZ	0.16 (*0.46*)

D

Method	MAD (*max*)
mPW1PW91/6-311+G(2d,p)*	0.10 (*0.27*)
revDSD-BLYP/cc-pVTZ	0.14 (*0.26*)
revTPSS/cc-pVTZ	0.10 (*0.37*)

*corrected

Figure 6. Mean absolute deviation (MAD) and maximum deviation (max) in ppm for selected DFs used to predict proton shifts of natural products. Absolute deviation in ppm for selected protons in stereocenters is shown. Corrected values were fitted against experimental data or adjusted with a scaling factor (for details, see Supporting Information).

set for mPW1PW91/6-311G(d), revTPSS/cc-pVTZ, and revDSD-BLYP/cc-pVTZ. The results are practically unchanged (see Supporting Information, Table S5), with MAD and SD differing by at most 0.01 ppm. It is concluded that the choice of reference compound does not influence the results.

It is common to calculate chemical shifts including solvation[16,91] or in the gas phase.[92,93] The effect of including implicit solvation within the conductor-like polarizable continuum method[78,94] was evaluated in comparison to gas-phase results for a small set of methods (Supporting Information, Table S7). Interestingly, with all DHDF methods (DSD-PBEP86, revDOD-PBEP86, and revDSD-BLYP) the gas-phase results yielded smaller MADs than those using CPCM ($CHCl_3$) solvation. In contrast, mPW1PW91 and revTPSS performed better with the inclusion of CPCM solvation, as might be expected for comparison with solution-phase experimental results. Nonetheless, the differences are relatively smaller and do not affect the overall results.

We also considered the level of theory utilized for geometry optimizations with a focus on the best-performing methods for ^{1}H NMR prediction, with the same method being employed for both the geometry optimization (gas-phase) and NMR calculations (CPCM solvation). All geometries were reoptimized separately with revDSD-BLYP/cc-pVTZ and revTPSS/cc-pVTZ followed by chemical shielding. Using revTPSS geometries and chemical shifts reduced the MAD and max values (MAD, 0.08 ppm; max, 0.24 ppm) compared to revTPSS calculations at the M06-2*X*/6-311+G(2d,p) geometries (MAD, 0.10 ppm; max, 0.26 ppm). It is apparent from the results in Figure 5 that revDSD-BLYP geometries lead to poorer chemical shifts with all methods.

For a final test, several natural products were considered as the complex structures provide real-world, challenging cases for ^{1}H NMR shift prediction. A set of natural products were selected that had previously been subjected to DFT studies to predict chemical shifts, which were recomputed with revTPSS and revDSD-BLYP as the best overall DF and DHDF, respectively. The selected structures were (A) rauvomine B ($CDCl_3$),[95] (B) aleutianamine (DMSO-d_6),[96] (C) eurotticin A (acetone-d_6),[97] and (D) aquatolide ($CDCl_3$)[98] that contain between 36 and 45 atoms and multiple stereocenters. All selected structures are based on X-ray structures (A, C, and D) or extensive NMR analysis (B). In each case, the calculated proton shifts (n = 52) are taken from Boltzmann-weighted conformations, with equivalent proton chemical shifts averaged (Figure 6). For all structures we also quote previously reported DFT results (B and D) or results calculated with the same method that was applied to ^{13}C chemical shifts (A and C). Invariably, the computational approach utilized a popular density functional, i.e., B3LYP or mPW1PW91, together with a Pople basis set of triple-ζ quality. For rauvomine B (A), Zeng et al. reported ^{13}C chemical shifts at the B3LYP/6-311G(d,p) level of theory in the gas phase;[95] here, we computed ^{1}H chemical shifts at the same level of theory with the CPCM implicit solvation model. Similarly, for eurotticin A (C) we applied the same level of theory used to compute ^{13}C for all four conformers,[97] while replacing the PCM implicit solvation model used by Zhong et al. by the CPCM model. The set of natural products includes solvents other than chloroform that had been consistently used thus far in this study. Individual results are provided in the Supporting Information (Table S10).

The revTPSS/cc-pVTZ method, which performs very well against the NMRH148 data set, consistently delivers the best results, matching or outperforming even scaled results from popular methods as in the case of aleutianamine and aquatolide (Figure 6, gray boxes). Out of the total 52 proton shifts computed for the natural products, the popular mPW1PW91 and B3LYP methods are in closer agreement with experiment than DHDF revDSD-BLYP. In fact, only in the case of aleutianamine does revDSD-BLYP exhibit better predictions than the popular hybrid-GGA method in the original reference, but at a considerably higher computational cost. DHDF revDSD-BLYP required roughly 1.5 order of magnitude longer to complete for larger molecules when compared to revTPSS/cc-pVTZ (for examples, see Supporting Information) and demanded dramatically more memory. Cursory inspection of individual protons in stereocenters (n = 16) revealed a similar trend with revTPSS displaying best predictions for most cases (8) followed closely by popular hybrid-GGA methods (7) and only then the best evaluated DHDF revDSD-BLYP (2). In all, predictions for natural products follow the same trend of results shown for the data sets of small molecules.

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5. CONCLUSIONS

Development of robust algorithms and advances in computer hardware have democratized the use of DFT methods to compute electronic structure and predict chemical properties. Unarguably, the success of DFT is also attributed to the results obtained with popular methods despite the presence of error cancelation. Recent developments in the area have led to the introduction of DHDFs (rung 5). The poor performance of current DHDFs for spectroscopical properties has been previously reported.[99] DHDFs were initially unable to describe long-range excitations—an issue which was later resolved with the introduction of specially designed TD-DHDFs.[100]

DHDFs have been recommended over lower-rung DFs for the calculation of a number of properties. The results presented here indicate that ^{1}H NMR chemical shifts are an important exception. The present results confirm that users can obtain similar or better results using less expensive DFT methods, e.g., a hybrid DF such as B3LYP or mPW1PW91 (rung 4) together with a Pople basis set of double- or triple-ζ, which may be corrected by readily available scaling factors.[101] On the basis of a comprehensive analysis of proton shift deviations from experimental values, the revTPSS/cc-pVTZ is demonstrated to be a reliable and affordable DFT method to compute proton shifts. Despite being overlooked in previous assessment studies, revTPSS (rung 3) emerges as a highly recommended method for ^{1}H NMR calculations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.1c00604.

Coordinates for all computed structures (ZIP)

Data for all calculations (ZIP)

Detailed description of computational methods, compounds in NMRH33 and NMR148, and results of additional calculations (PDF)

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Notes

The authors declare no competing financial interest.

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